

Total No. of Questions : 8]

[Total No. of Printed Pages : 1

Roll No

EE-504 (GS)
B.E. V Semester Examination, June 2020
Grading System (GS)
Digital Electronics and Logic Design
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. The message below coded in the 7-bit Hamming code is transmitted through a noisy channel. Decode the message assuming that at most a single error occurred in each code word. 1001001, 0111001, 1110110, 0011011.
2. Write short note:
 - a) Diode as switch
 - b) Bipolar transistor as switch
3. Convert the following:
 - i) $(247.8)_{10} = (?)_2$
 - ii) $(247.8)_{10} = (?)_3$
 - iii) $(AF.16C)_{16} = (?)_8$
 - iv) $(4163)_8 + (7320)_8 = (?)_2$
 - v) $(18.6)_9 = (?)_{11}$
4. Convert the following binary numbers to decimal and to BCD
 - i) 11011101
 - ii) 10001110
5. Solve for X
 - i) $(F3A7C2)_{16} = (X)_{10}$
 - ii) $(2AC5)_{16} = (10949)_X$
 - iii) $(0.93)_{10} = (X)_8$
 - iv) $(4057.06)_8 = (X)_{10}$
6. Write short notes on any two of the following:
 - a) Maximum clock frequency
 - b) Level triggered and edge triggered flip-flop's
 - c) FPGA
7. A combinational circuits is defined by one function
$$F_1(A, B, C) = \sum m(0, 1, 3, 7)$$
$$F_2(A, B, C) = \sum m(1, 2, 5, 6)$$
Implement this circuit with PLA.

OR

Compare the characteristics of different logic families.

8. Write short note on any two of following:
 - a) Universal shift register
 - b) Up down counters
 - c) TTL based logic gates
 - d) D/A and A/D conversion
